

Lecture 17: Introduction to Paging

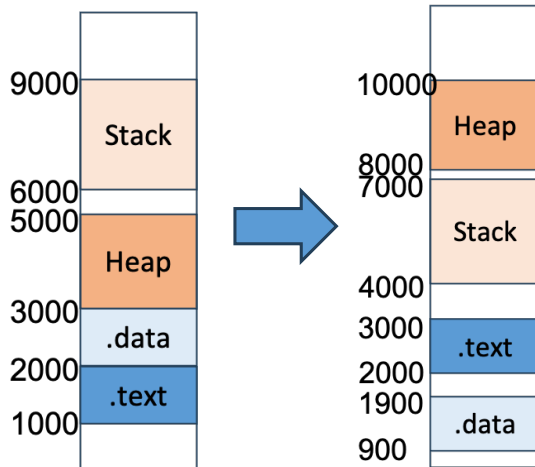
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Last Lecture



Base	Bound	Type	Prv
2000	3000	RE	U
4000	7000	RW	U
900	1900	RW	U

Global Descriptor Table (GDT)

Visible Part	Hidden Part	
GDT index	Base Address, Limit, Access Information	
		CS
		SS
		DS
		ES
		FS
		GS

Segment Selector	Offset
------------------	--------

- Segment wise base/bound enables non-contiguous inter-segment memory
- VA has segment selector bits to identify the segment and an offset into that segment
- Base and bound for each segment stored in GDT
 - Address of GDT stored in GDTR
- x86 platform supports six segment registers per CPU
 - VA segment selector helps in choosing segment register
 - Each segment register contains the offset into GDT
 - Index into GDT is sum of GDT base address inside GDTR and above offset
 - Offset inside VA added to the segment's base address obtained from GDT
 - Base/bound is cached inside segment register's hidden part
- Segment registers must be saved/restored at context switch

Today's Class

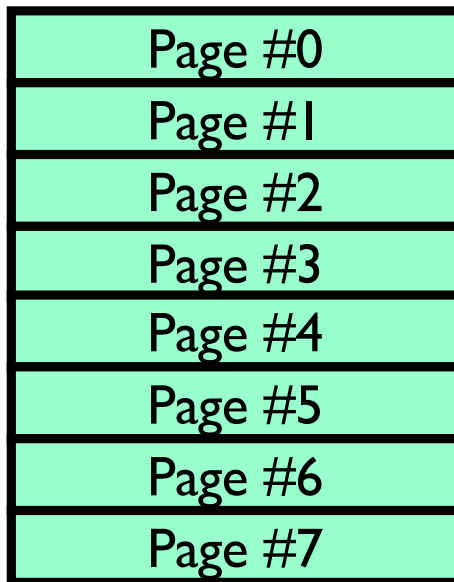
- Introduction to paging

Mapping Virtual to Physical Address

- Approaches

1. Virtual address is same as the physical address
2. Virtual address is not the same as physical address
 - a) Using base & bound for VA to PA mapping
 - b) Using segmentation for VA to PA mapping
 - c) **Paging**
 - a) **Using Page Table (PT) in RAM and having a base register for PT**

Paging



Physical Memory

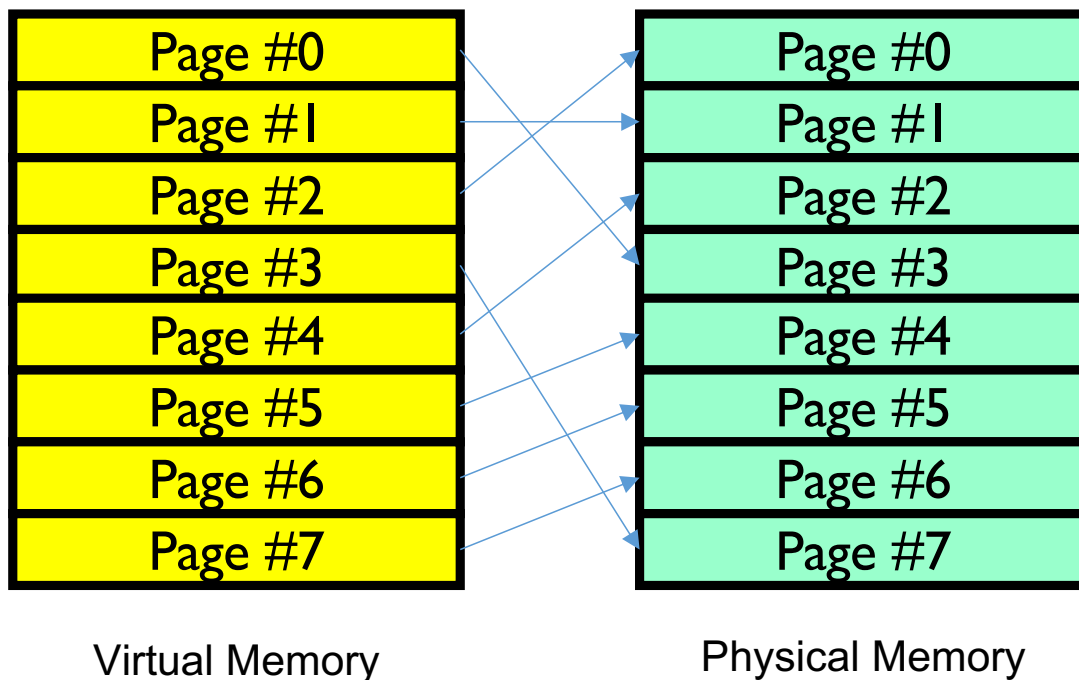
- Segmentation is easy to implement but it has major drawbacks because segments could have variable sized memory allocation
 - Can lead to **external fragmentation**
 - To allocate space for a new process, segments of some other process has to be swapped to the disk (huge overhead)
- Paging solves both the above issues by dividing the available memory into small and fixed size blocks (pages)
 - The memory can then be viewed as an array of fixed sized slots
 - For example, with a page size of 8 bytes, we can have an array of 8 pages on a DRAM whose size is 64 bytes

Total Number of Pages

- Default page size on Linux is 4096 bytes (4KB)
- Pages on physical memory depends on DRAM size
 - Assume DRAM size is 64GB, then total number of pages in **physical memory** will be $64\text{GB}/4\text{KB}$, i.e., 16×10^6 pages
- Pages in virtual memory is fixed as per the addressing mode
 - Total number of pages in **32-bit** addressing mode is $2^{32}/4\text{KB}$, i.e., 2^{20}
 - Total number of pages in **64-bit** addressing mode is $2^{64}/4\text{KB}$, i.e., 2^{52}

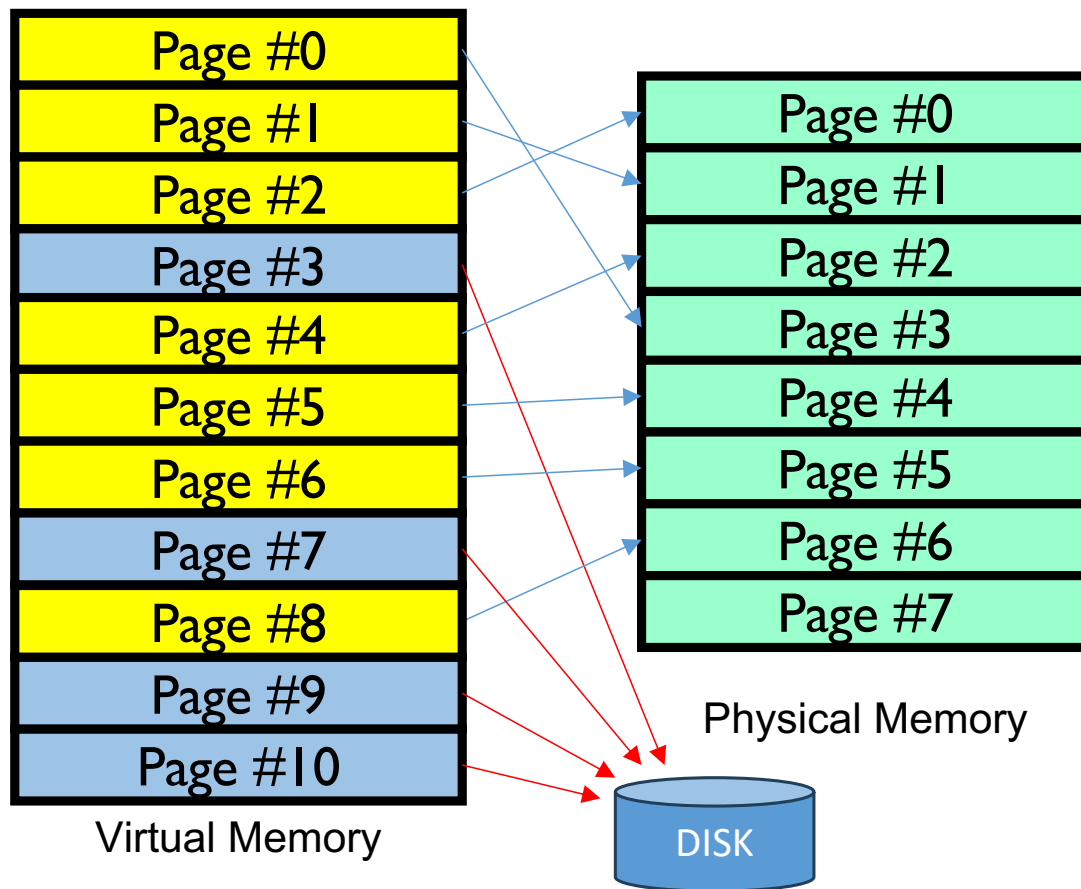
If you think of a 32-bit address space as the size of a tennis court, a 64-bit address space is about the size of Europe(!) -- OSTEP Book

Paging



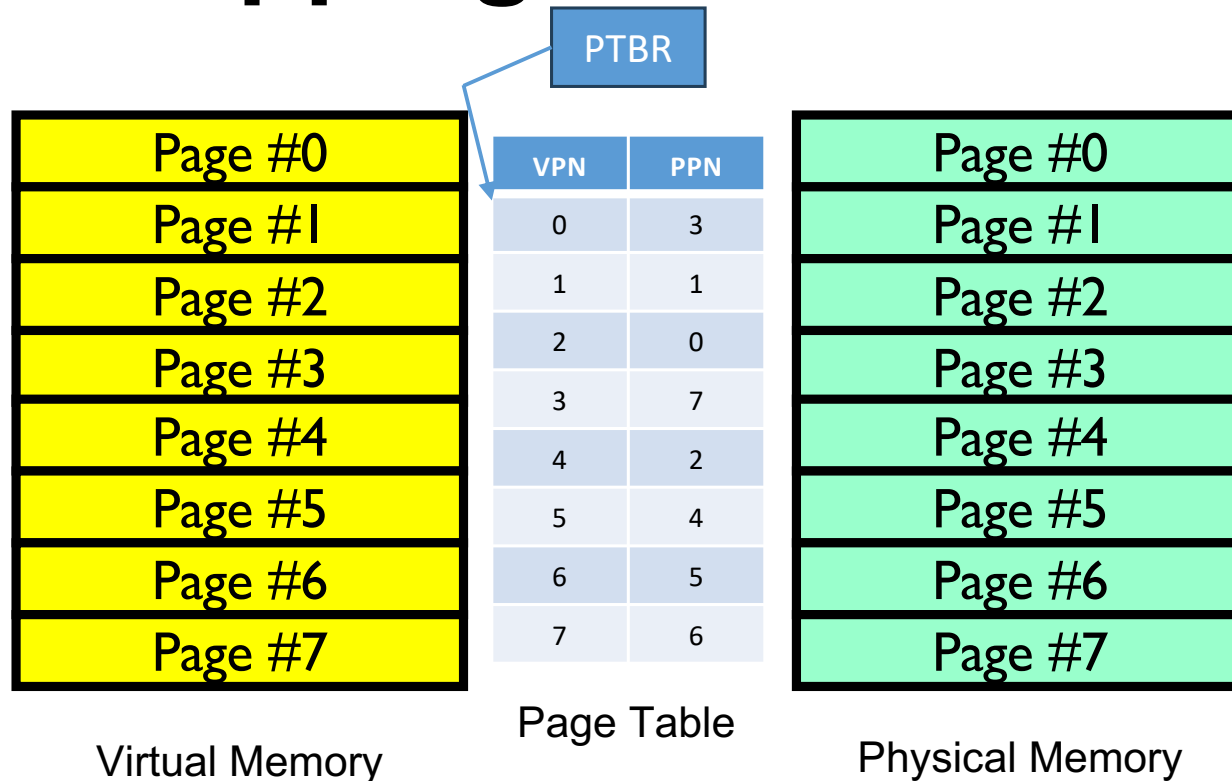
- Process's segments would still be of variable size, but now instead of a contiguous chunk of memory, they could have **non-contiguous** mapping with physical pages (both inter/intra-segment)
- **How about internal fragmentation?**
 - Yes, if a segment requires 5000 bytes of memory, it would have 2 page allocations where 4092 bytes would forever remain free in the second page

Paging



- Swapping out to disk can still happen, but which one would have minimal overheads?
 - Segmentation Or paging?
- Lesser overheads in paging as compared to segmentation
 - Instead of swapping out an entire segment, only unused pages of program segment(s) will be swapped out
- Any issue still remaining?
 - How to map Virtual Page Number (VPN) to Physical Page Number (PPN)?

Mapping VPN to PPN



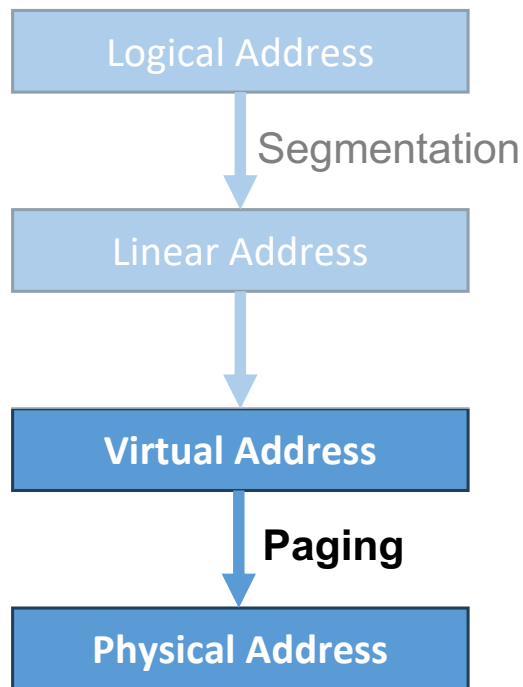
- Mapping of VPN to PPN is carried out using a data structure called as Page Table
- It is stored in DRAM
- Each CPU has a Page Table Base Register (PTBR) that holds the base address and Page Table Length Register (PTLR) that holds the size of the page table
 - PTBR help fast retrieval of base address of PT
 - PTLR helps in performing bounds check for virtual page (e.g., in the PT shown here, accessing VPN=10 will generate exception as PTLR=8)

A System Using Paging

```
void scheduler() {
    while(true) {
        lock(process_table);
        foreach(Process p: scheduling_algorithm(process_table)) {
            if(p->state != READY) {
                continue;
            }
            p->state = RUNNING;
            unlock(process_table);
            swtch_pagetable_base_registers(scheduler_process, p);
            swtch(scheduler_process, p);
            // p is done for now..
            lock(process_table);
        }
        unlock(process_table);
    }
}
```

- Page table resides in memory
- Process's PCB has a field to store PTBR+PTLR
- PTBR+PTLR of a running process is saved during context switch

Segmentation Along with Paging

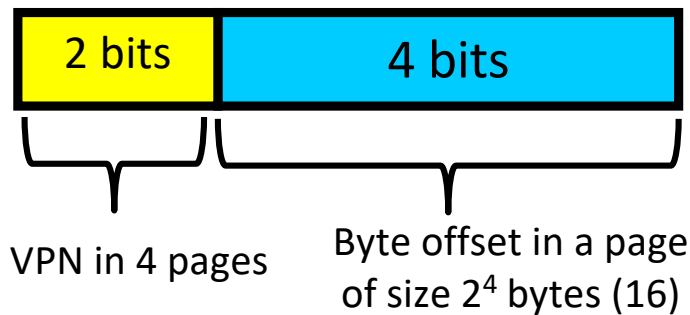


- Segmentation cannot be disabled in x86 processors, but address translation is complex using the segmentation-based approach
 - Hence, to simplify the address translation, segment registers are set to have the same base (0) and bound ($2^{32}-1$) values (Lecture #16)
- The address visible to a programmer is actually called as Logical address, as it has an associated segment selector (Lecture #16)
 - Segmentation converts this logical address to linear address, which is then converted into virtual address to support paging
- However, as the base and bound of each segment registers maps to entire 2^{32} memory size (4GB), logical address will be the same as virtual address
- Hence, we will call the address visible to the programmer as a virtual address in all the remaining slides/lectures

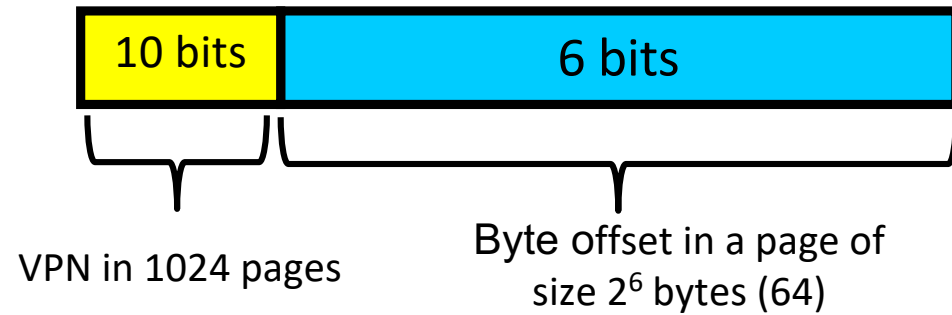
Encoding VPN inside Virtual Address

- How many VAs can reside in a 4KB page size?
 - $4096 \text{ bytes} / 4 \text{ bytes} = 1024$ virtual addresses
- How to find the VPN for a given VA?
 - Encode (prefix) the VPN inside each VA (**next slide**)
- How many bits required to represent a VPN for pages of N bytes in a virtual memory of M bytes (total M/N pages)?
 - $M=64$ (2^6) and $N=16$
 - Pages = 4 and bits required to identify 4 pages are 2 ($2^2=4$)
 - $M=4\text{GB}$ (2^{32}) and $N=4\text{KB}$
 - Pages = $2^{32}/4096 = 2^{20}$ and bits required to identify 2^{20} pages are 20

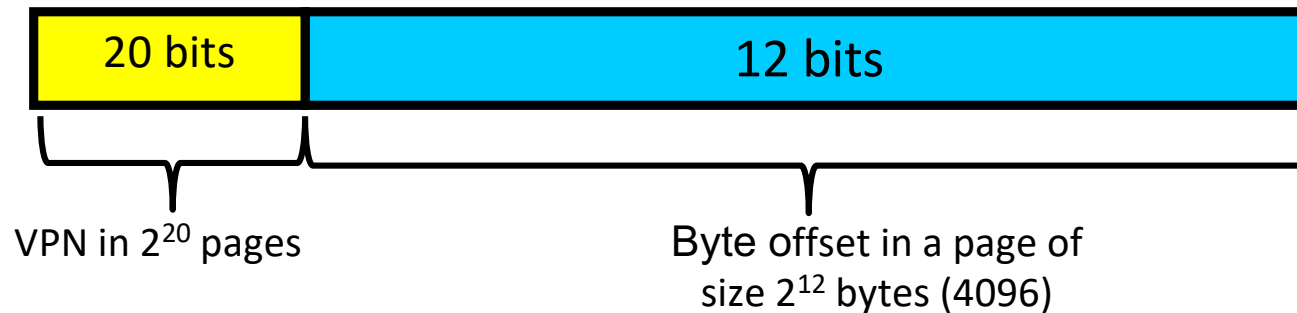
Two Components of a Virtual Address



6-bit addressing with page size of 16 bytes

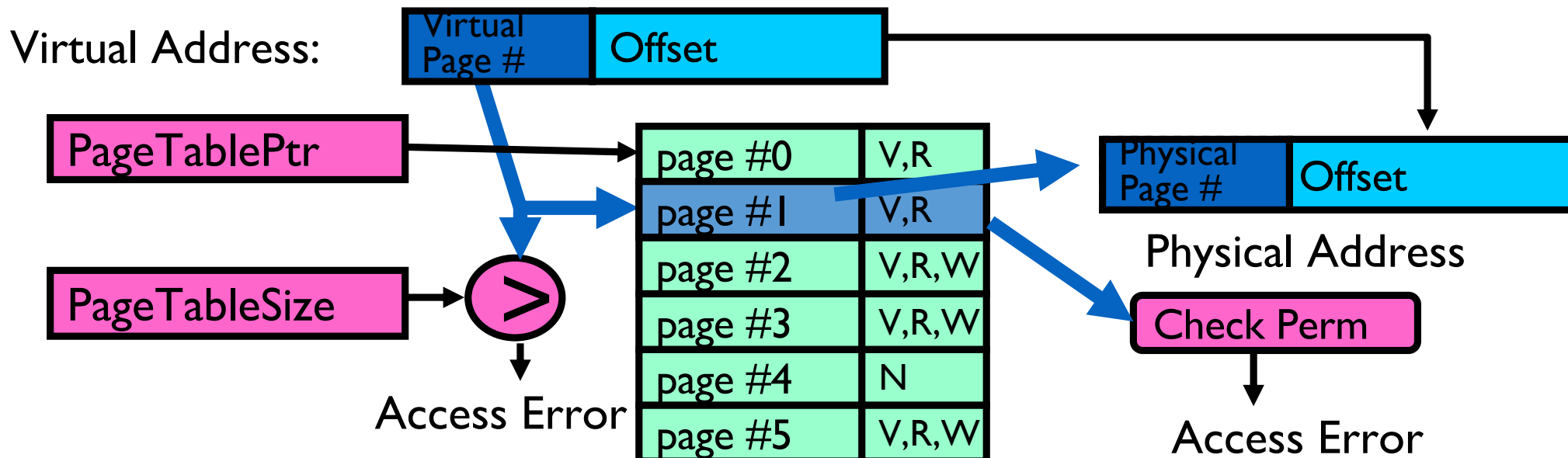


16-bit addressing with page size of 64 bytes



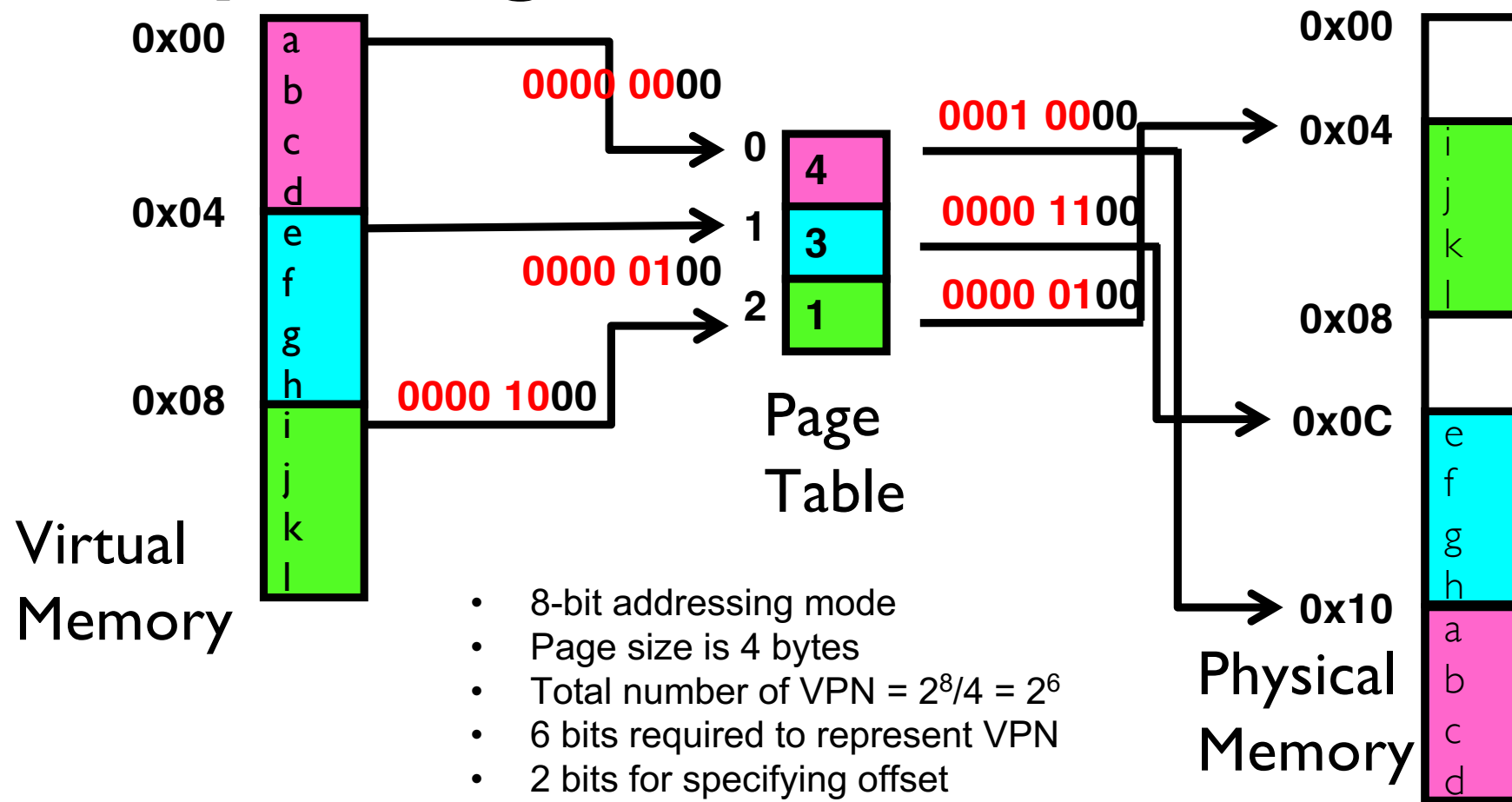
32-bit addressing with page size of 4KB

How to Implement Simple Paging?

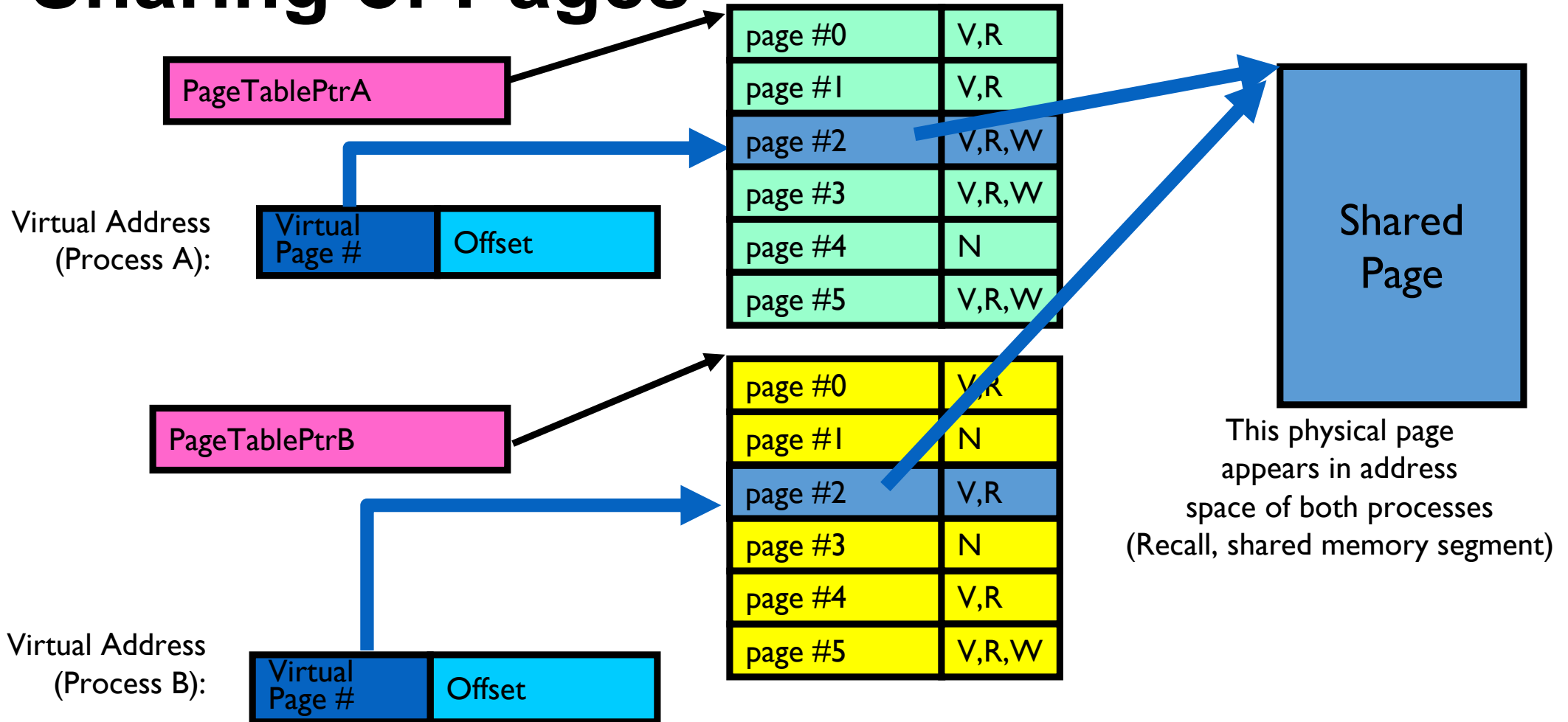


- Page Table (One per process and resides in physical memory)
 - Contains physical page number and permission for each virtual page (e.g. **V**alid bits, **R**ead, **W**rite, etc)
- Virtual address mapping
 - Offset from Virtual address copied to Physical Address
 - Virtual page # is all remaining bits (Physical page # copied from table into physical address)
 - Check Page Table bounds and permissions

A Simple Page Table in Action

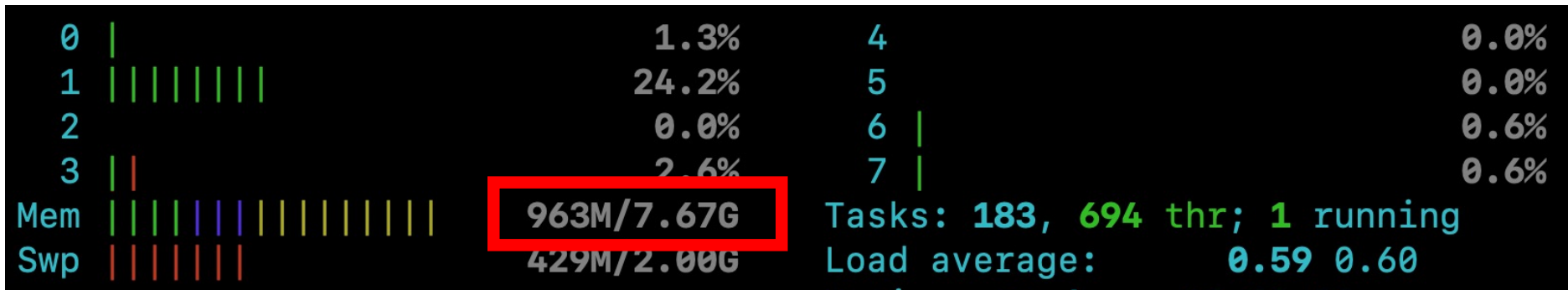


Sharing of Pages



Physical Page Allocation in Linux

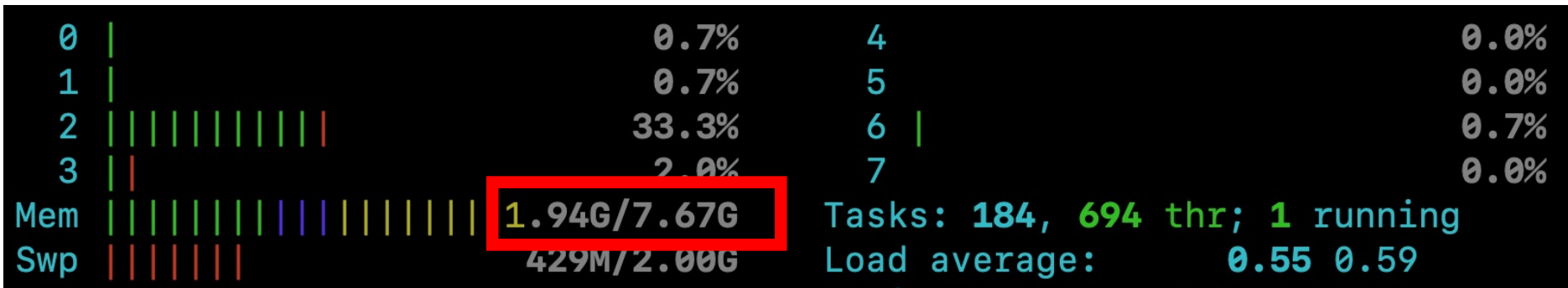
```
char * array = (char*) malloc(sizeof(char)*1024*1024*1024);
```



- The above program allocates 1GB memory using malloc, but output of **htop** command doesn't show 1GB memory usage!
 - I did not see any change to that memory even after the above program finished, which implies its default memory usage in my system
 - What is happening?

Physical Page Allocation in Linux

```
char * array = (char*) malloc(sizeof(char)*1024*1024*1024);
memset(array, 0x00, sizeof(char)*1024*1024*1024);
```

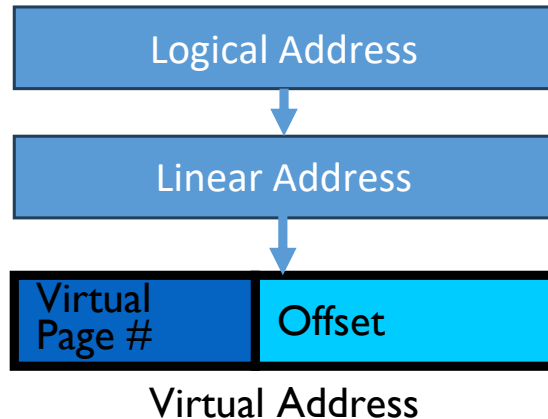


- Just writing to that allocated space changes the output of **htop** command to show the 1.94GB memory usage (1GB from the program and 0.94GB of default memory)
 - What just happened now?

Physical Page Allocation in Linux

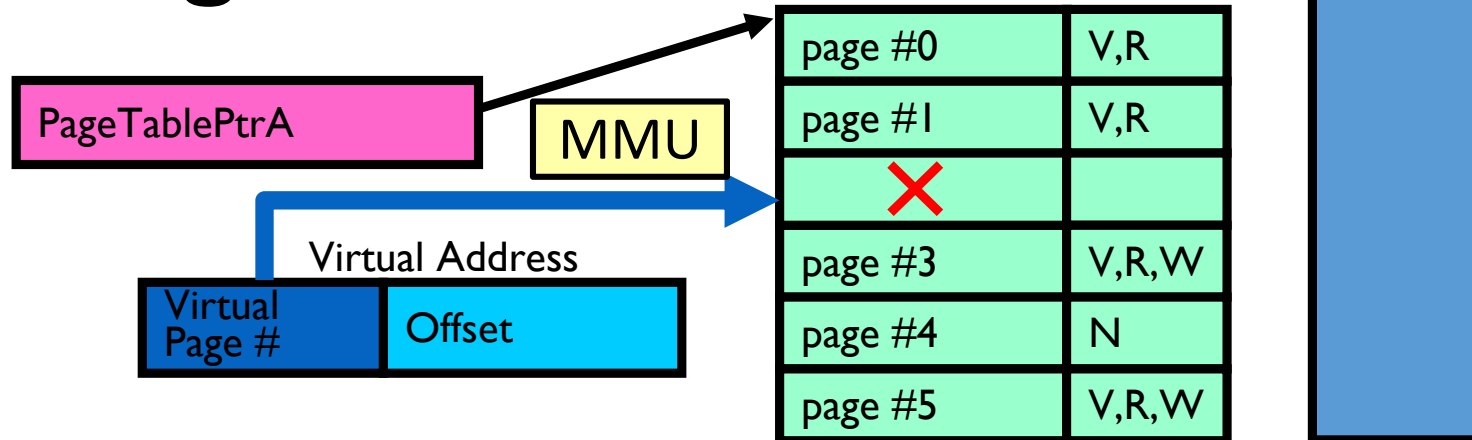
- Linux uses lazy memory allocation policy to save memory space (RAM is precious!)
 - Not just for malloc, but even while loading program segments
 - Physical memory for the segments are not allocated until accessed
 - E.g., if a program has a .data segment, but the user doesn't access the global variables (residing in .data segment) then it doesn't make sense to allocate memory for .data segment upfront (load time)
- So, how the physical memory is allocated?
 - During the program execution, when the program attempts to **read/write** to that memory
 - Linux uses a mechanism known as **Page Fault** for lazy page allocation

Page Fault in Action



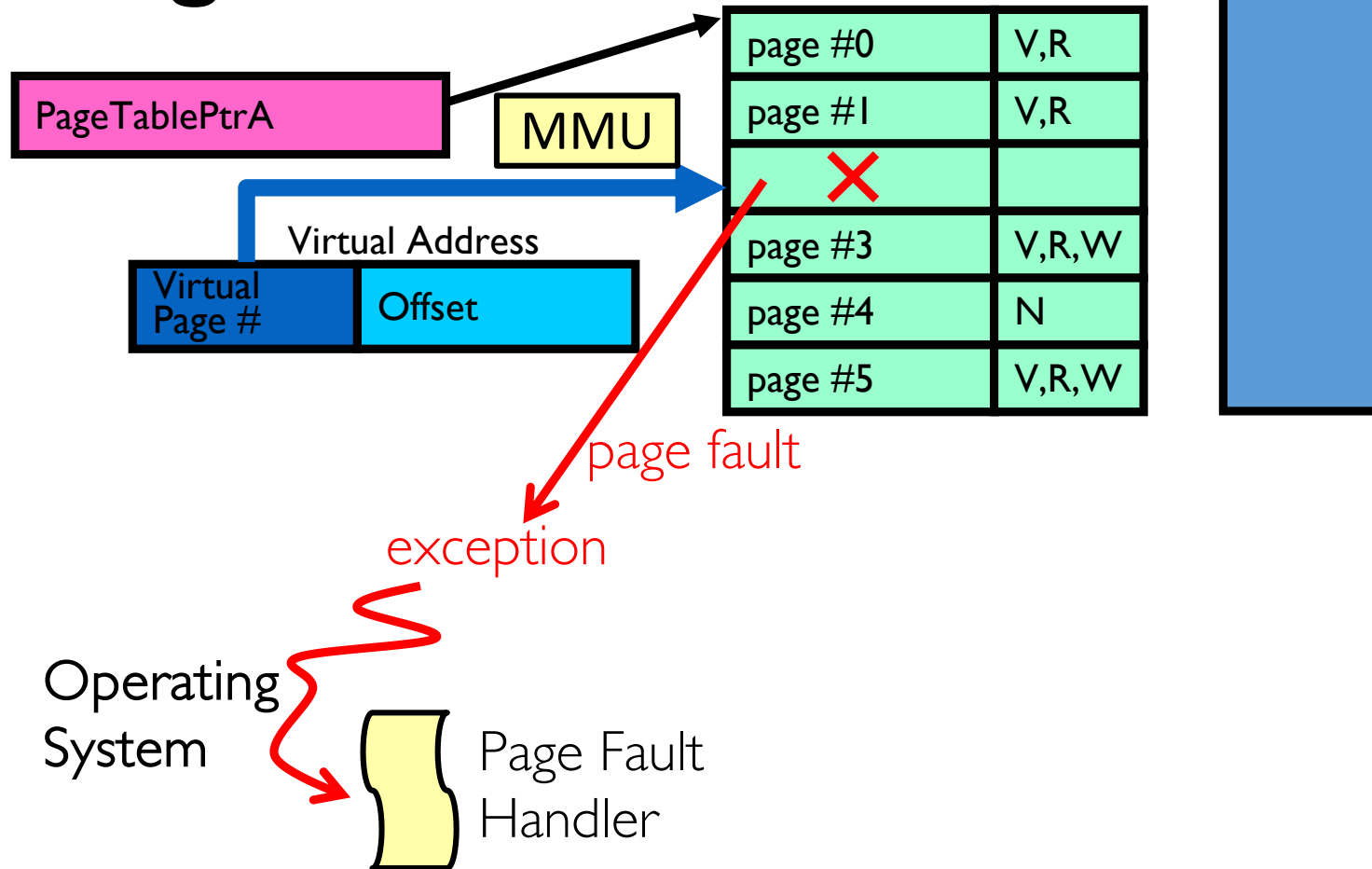
- Program attempts to access an address
- Segmentation converts this logical address to linear address, which is then converted into virtual address to support paging

Page Fault in Action



- This address is valid (i.e., it points to a valid heap allocation)
- However, the page table doesn't have any entry for this VA to PA mapping

Page Fault in Action

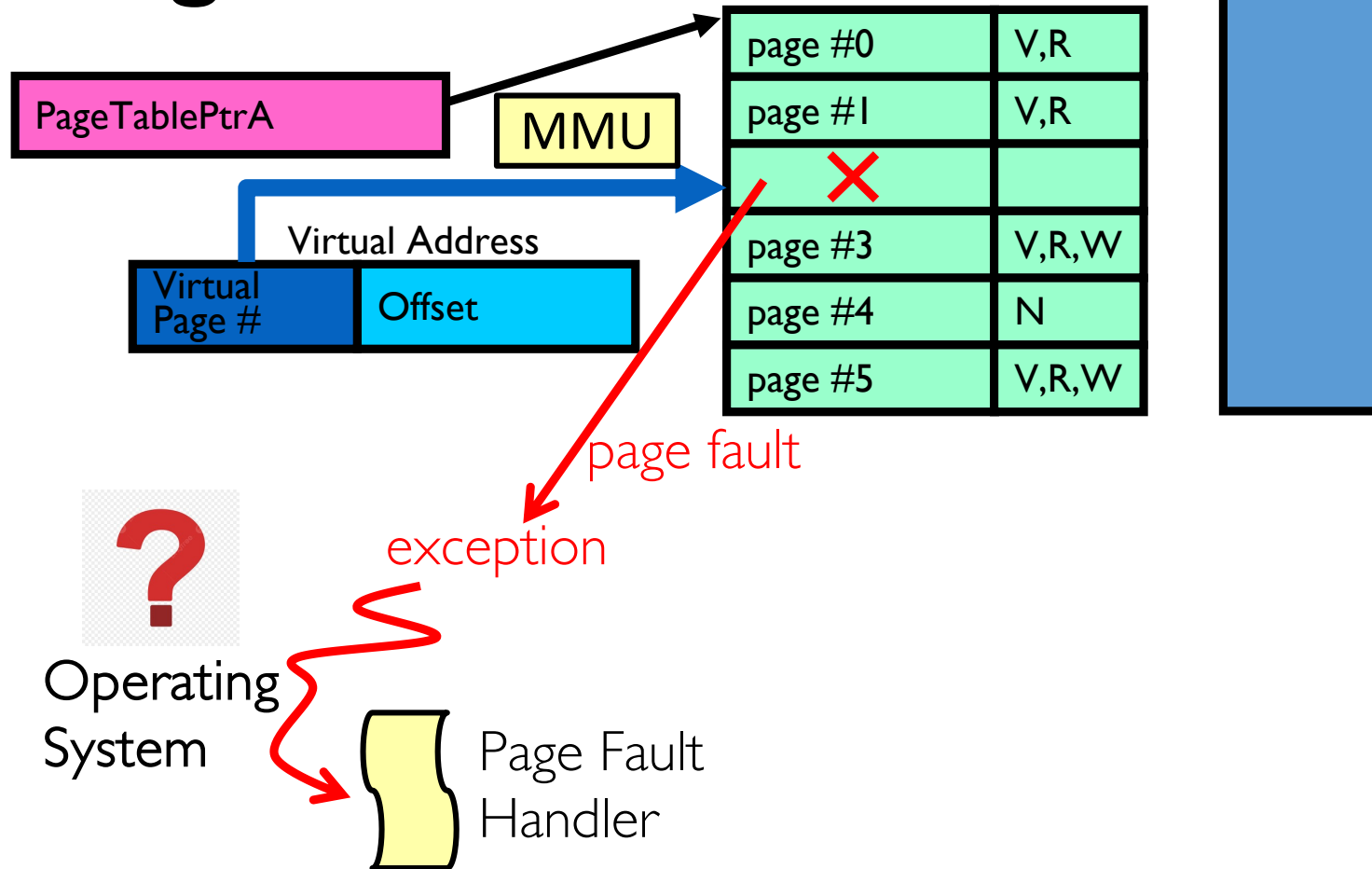


DRAM



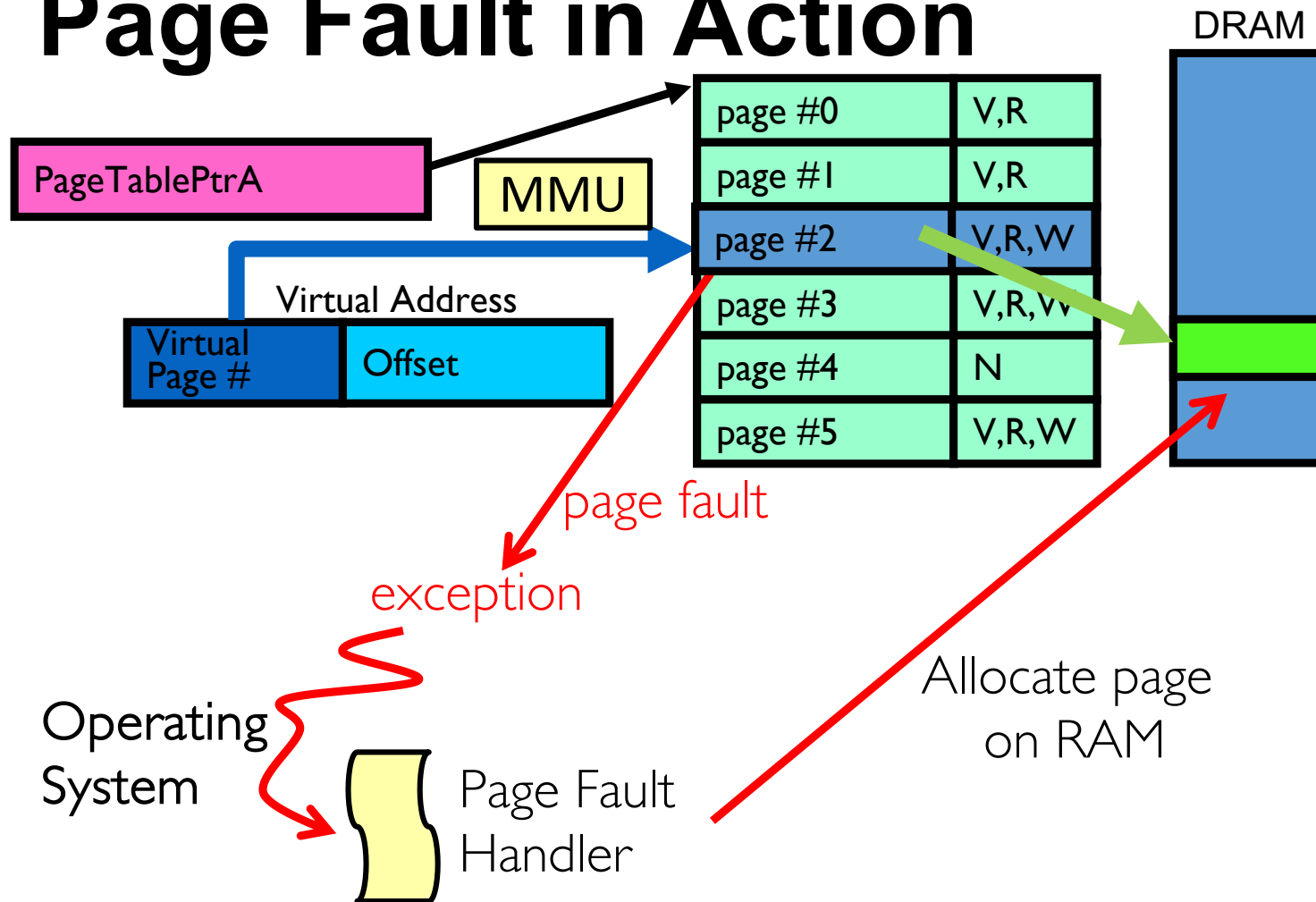
- Program state is saved and Interrupt 14 is generated by the MMU indicating page fault exception

Page Fault in Action



- OS will decide what action is to be taken
 - It could be that the page was allocated but swapped out to disk (i.e., demand paging, we will see in future lecture)
- OS finds out that no allocation was ever made

Page Fault in Action



- OS will allocate page(s) on DRAM
- Page table entry is updated
- Program execution (state) is restored from the same location that caused the page fault

Reference

- Chapter 18 in OSTEP book (“Paging Introduction”)

Next Lecture

- Translation Lookaside Buffer (TLB)